GUSTAVO BAQUERO SANCHEZ

Junior Mechanical Engineering Student

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EDUCATION

Northern Arizona University, Honors College 08/2024 – 05/2028

B.S. Mechanical Engineering, Junior

GPA 3.4 · NSF S-STEM Scholar · National Merit Scholar · NAU Honors Scholar

EXPERIENCE

Undergraduate Researcher, Funded Rocketry Research (LatticeShield), NAU 08/2026 – Present

- Selected ASA Aero over PLA-CF, ABS-CF, and mandrel-wrapped composites after tensile testing (Mark-10 30ES), hoop testing (shear strain, expansion, pressure), and flame exposure, for strength-to-weight and temperature resistance.
- Replicated those strength tests in ANSYS on smaller rocket components, reaching 85–91% agreement with measured results on all core components.
- Cut aerostructure mass 25% (25 g) with gradient gyroid lattices in nTopology and SOLIDWORKS, verified by weighing the original and lightweighted prints.
- Designed a composite fin can (CFPEEK lattice, carbon fiber, ceramic coating) and verified it against high-speed aerodynamic shear in SOLIDWORKS and ANSYS.
- Added 1–2° fin canting for passive spin stabilization, avoiding an estimated 30–40 g of active-stabilization hardware.

Founder, Federal Subcontracting (DLA DIBBS) 09/2025 – Present

- Registered DoD subcontractor (CAGE 208T1; SAM.gov active through 05/2027). Submitted 13 bids, won 3 contracts, and delivered 3D-printed parts for obsolete vehicles.
- Built a workflow that screens solicitations for eligibility (non-ITAR, no clearance required, CMMC risk), saving ~3 hours per batch.

Lab Manager, IDEA Lab (Fabrication Facility), NAU 09/2025 – Present

- Run the fabrication facility 20 hrs/week and supervise 3 student operators.
- Manufactured final parts for 20+ capstone teams (laser-cut aircraft components, 3D-printed hydroelectric turbines), checking tolerance, fit, and finish on every part.
- Advise teams on tolerances, fits, and material and process selection across FDM, Stratasys, metal powder-bed fusion, and laser cutting.
- Maintain and troubleshoot the printers, and design custom replacement parts.

National Competitions Manager, ASME, NAU 05/2025 – Present

- Raised ~$6,000 for travel and parts, returning NAU to e-HPVC and IAM3D for the first time in 10 years.
- Handled legal paperwork and workspace for student teams.
- e-HPVC: secured sponsors, including Nova Kinetics, for build materials and space-grade frame welding, and designed the outer fairing the team adopted.
- IAM3D Mars Rover: designed and manufactured rover parts, including the chassis.

SKILLS

Design & Analysis: nTopology, SOLIDWORKS, ANSYS, Fusion 360, MATLAB, RocketSim

Testing: Mark-10 30ES force gauge, hoop tensile testing, strain measurement

Manufacturing: Stratasys Fortus 450mc/250mc and J35, Concept Laser MLab Cusing (metal powder-bed fusion), Bambu, Creality, Flashforge, MakerBot, Ultimaker

Languages: English (native), Spanish (native)

ACTIVITIES & CERTIFICATIONS

ASME (Officer) · SHPE · AAAS · OSHA 10-Hour